

Informal (Field-Based) Learning

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Abstract

Most learning in the workplace occurs outside of formal learning environments—it happens informally, “in the field.” In this review, we share what is known about how such informal field-based learning (IFBL) works, offer guidance on how to promote healthy IFBL, and identify future research needs. We first situate IFBL within the broader stream of the learning literature. On the basis of the literature and organizational needs, we propose a CAM-OS framework that emphasizes five personal and situational readiness factors for enabling constructive IFBL: Capability, Awareness, Motivation, Opportunity, and Support. We use the framework to offer practical, evidence-based advice for each of three stakeholder groups—senior leaders, managers, and employees—and conclude with suggested avenues for future research. The review is grounded in the research literature with an emphasis on implications for practice.

INTRODUCTION

To remain competitive in today's dynamic work environment, organizations need to continually enhance their employees' competencies and capabilities. Traditionally, organizations have attempted to do this by investing primarily in formal training and development (T&D) activities. For example, in 2018, US organizations spent an estimated \$204 billion on formal T&D (ATD 2019). Although T&D can play a critical role in improving individual, team, and organizational performance (Bell et al. 2017), a substantial proportion of workplace learning happens informally. Estimates have ranged from 70% (Bear et al. 2008) to more than 90% of competency acquisition occurring through informal means (Flynn et al. 2006, Tannenbaum 1997). Although the specific percentages may be debatable, clearly employees perceive that the majority of their learning happens outside of formal T&D, with many reporting that informal learning is a daily experience (De Grip 2015).

As organizations began to acknowledge that informal learning plays a key role in remaining competitive (Roy 2010), research on the topic increased as well. **Figure 1** shows a frequency graph of informal learning articles published from 1970 to 2020. Notably, there has been growing interest in informal learning, particularly beginning in the early 2000s and accelerating exponentially since. One potential reason for this growth is the idea that IFBL can be a low-cost alternative to or used in conjunction with formal T&D (Wolfson et al. 2018). Although research has shown that informal learning is clearly valuable (Cerasoli et al. 2018), and we can offer useful advice based on that research, as a field we are just beginning to learn how it works and what factors contribute to its success. Additional research needs to be conducted to further understand the antecedents, consequences, and enabling factors facilitating individuals' engagement in and ultimate benefit from IFBL (Wolfson et al. 2018).

INTENT AND SCOPE OF THIS REVIEW

In this review, we focus on informal field-based learning (IFBL), which is defined as “engaging in intentional, self-directed behavior aimed at learning new, work oriented, and organizationally valued content outside of a formal learning program” (Wolfson et al. 2018, p. 16). We share what is known about how IFBL works, offer guidance on how to promote healthy IFBL, and identify where future research would be most beneficial. To accomplish this, we first define IFBL and

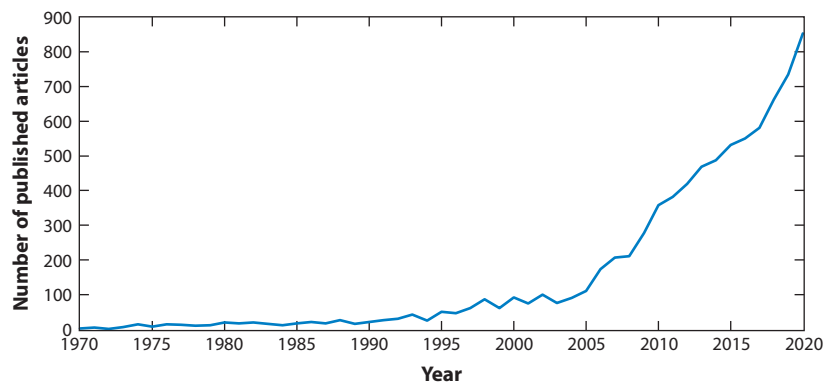


Figure 1

Frequency graph of informal learning articles from 1970 through 2020 showing growing interest in informal learning, particularly beginning in the early 2000s and accelerating exponentially since.

explain how it fits in the broader stream of the learning literature, using a three-dimensional taxonomy that situates IFBL among related constructs. Second, we propose a CAM-OS framework that is based on the literature and incorporates common organizational dynamics. The framework emphasizes five factors (Capability, Awareness, Motivation, Opportunity, and Support) that are potential leverage points for enabling effective IFBL. In describing the framework, we highlight representative research, expanding on recent reviews of learning (e.g., Noe et al. 2014), but our review is intentionally selective rather than comprehensive. Third, we consider IFBL through the lens of three key stakeholder groups—senior leaders, managers, and employees—and use the CAM-OS framework to offer practical, evidence-based advice for each group. Finally, we suggest avenues for future research. The review is grounded in the research literature with an emphasis on implications for practice.

WHAT IS INFORMAL FIELD-BASED LEARNING AND WHY IS IT IMPORTANT?

To appreciate the nature and value of IFBL, we should understand where it fits in the broader scheme of learning and T&D. As highlighted in many recent articles, a plethora of related constructs have been introduced in this space (e.g., Aguinis & Kraiger 2009, Noe et al. 2014, Wolfson et al. 2018). Wolfson et al. (2018) proposed a subset of informal learning that they termed IFBL and defined as “engaging in intentional, self-directed behavior aimed at learning new, work oriented, and organizationally valued content outside of a formal learning program” (p. 16). They proposed that IFBL consists of three subdimensions or types of learning behaviors: (a) seeking feedback and reflection-based learning, (b) vicarious learning, and (c) learning through experimentation and new experiences.

As part of feedback and reflection-based learning, individuals may actively seek feedback from others; seek and receive coaching or advice from experts; and think about, debrief, or discuss on-the-job experiences. Vicarious learning can occur by intentionally observing others doing their job, asking questions of experts, or having someone demonstrate how to perform a task. Finally, learning through experimentation and new experiences can occur by performing a task in novel ways; actively seeking and experiencing novel assignments, tasks, or situations; or through trial-and-error approaches in an attempt to find new or improved solutions. In this review, we adopt Wolfson et al.’s (2018) definition of IFBL and their categorization of IFBL behaviors.

How does this view of IFBL fit into the larger construct space? Wolfson et al. (2018) described various types of learning constructs as occurring across three continua: (a) informal–formal, (b) intentional–incidental, and (c) self-directed–other-directed. To reconcile IFBL with related constructs, we advance an organizing framework that builds on Cerasoli et al. (2014) and Wolfson et al. (2018) and considers the various ways researchers have discussed learning across those dimensions. In **Figure 2**, we offer a visual depiction situating various learning constructs in a three-dimensional space, with definitions presented in the Appendix.

Formal Versus Informal

The first continuum considers learning that ranges from formal to informal, with formal learning having greater structure and defined consistency, for example, by pre-establishing learning goals, learning experiences, and timing. Training resides on the formal end of the continuum and is intended to serve the organization strategically through an emphasis on the acquisition of knowledge, skills, abilities, and other characteristics (KSAOs) that can boost organizational effectiveness (Kraiger & Ford 2006). Other constructs, such as on-the-job training (Rothwell &

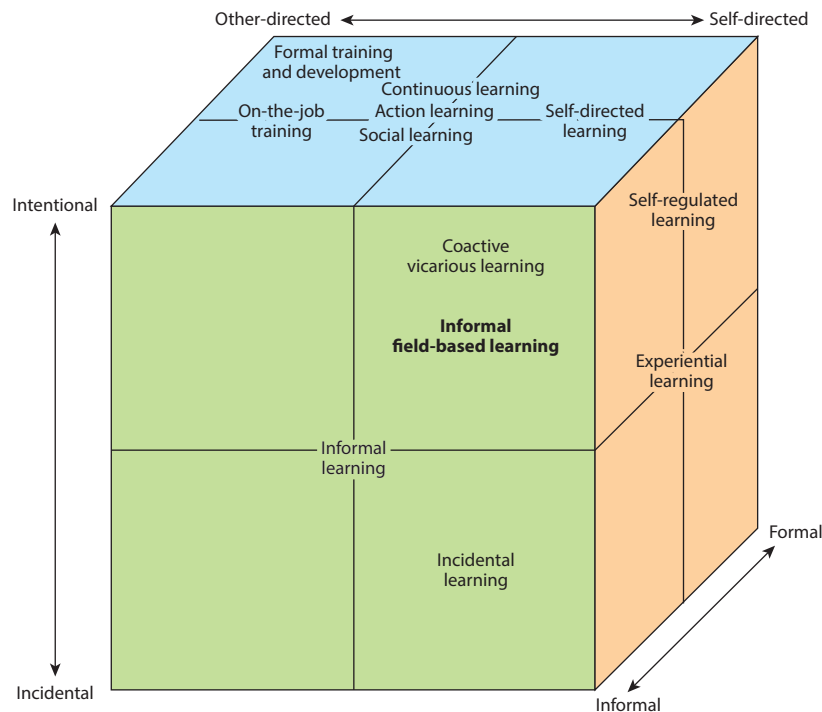


Figure 2

Visual depiction situating various learning constructs in a three-dimensional learning space. Learning types vary along the dimensions of formal to informal, other-directed to self-directed, and incidental to intentional, with informal field-based learning being informal, self-directed, and intentional. Figure adapted from Cerasoli et al. (2014) and Wolfson et al. (2018).

Kazanas 1994), action learning (Yorks et al. 1999), continuous learning (London & Sessa 2006), self-directed learning (Clardy 2000), self-regulated learning (Sitzmann & Ely 2011), and experiential learning (Bandura 1962), may incorporate aspects of formal and informal learning, and thus reside toward the middle of this continuum. For example, continuous learning can be stimulated by T&D activities (more formal) or by simply observing others (more informal).

Other-Directed Versus Self-Directed

Along the other- versus self-directed learning continuum, learning differs in terms of who structures and directs the learning. Formal T&D as well as on-the-job training tend to be other-directed, as supervisors or an organization's learning and development function establish the content and delivery of the learning material. Continuous learning, action learning, and informal learning (Marsick et al. 1999) tend to traverse the other- versus self-directed continuum. For example, with informal learning, an individual could engage in observation at their own discretion or might be offered an opportunity to observe someone. Toward the self-directed end of the continuum there is also self-regulated learning, experiential learning, and coactive vicarious learning. Coactive vicarious learning is a "discursive learning process where individuals (i.e., a model and learner) intentionally share and jointly process a model's work experience(s) in interpersonal interactions to construct an emergent, situated understanding of the experience(s)" (Myers 2018, pp. 613–614).

Intentional Versus Incidental

The final continuum ranges from intentional to incidental, where learning differs in the intentionality or serendipity with which it occurs. Incidental learning can be viewed as a subset of informal learning but is outside our definition of IFBL. Incidental learning generally happens without forethought and at times without awareness, as a by-product of other activities (Watkins & Marsick 1992). Experiential learning, workplace learning, and informal learning can often have both intentional and incidental elements to them, as an event that triggers learning may occur intentionally or not. Several of the other learning constructs that we alluded to previously fall on the intentional end of the continuum, including social learning (Bandura 1962), continuous learning, action learning, formal T&D, on-the-job training, self-directed learning, self-regulated learning, and coactive vicarious learning.

IFBL, as Wolfson et al. (2018) defined it, resides toward the informal, self-directed, and intentional ends of their respective continua. The inclusion of intentionality differentiates it from some other types of informal learning that can have a more incidental or even unconscious aspect to them (Tannenbaum et al. 2010). We have adopted this focus because intentionality allows individuals and organizations alike to be more purposeful in their approach to informal learning, rather than hoping it occurs serendipitously. Stated differently, viewing IFBL as an intentional, self-directed, informal way of learning acknowledges the active, conscious role an employee takes in much of their learning, while holding open the potential for the organization to support, encourage, and enable employees to do so more effectively. It suggests that it is possible to promote constructive IFBL without exhibiting the same degree of organizational control associated with formal T&D experiences. We next turn to our organizing framework of IFBL that highlights how situational and personal factors influence who engages in IFBL behaviors and whether those behaviors are fruitful.

CAM-OS: AN ORGANIZING FRAMEWORK OF INFORMAL FIELD-BASED LEARNING

We developed the CAM-OS framework, depicted in **Figure 3**, to organize and describe what is currently known about IFBL, including various antecedents or prerequisites that can enable IFBL behaviors and desired outcomes. The framework also clarifies where actions can be taken to enhance constructive IFBL. The model incorporates the three main types of IFBL behaviors along with several common, desirable outcomes (shown in green). Some of the outcomes clearly benefit the learner, such as learning, career progression, and affect (e.g., improved job satisfaction); some can provide a benefit to the organization, such as enhanced performance and a better talent pipeline. But not all IFBL behaviors lead to these constructive outcomes. Although the consequences of IFBL are usually positive, at times unforeseen missteps may occur. Consider, for example, an employee who decides to informally try a new technique that creates a safety hazard for his coworkers.

The extent to which employees engage in IFBL behaviors and whether those behaviors yield desired outcomes is a function of both personal readiness (Capability, Awareness, and Motivation) and situational readiness (Opportunity and Support). Understanding these five actionable readiness factors (shown in orange) can help clarify where and how to promote constructive IFBL. In addition, the research suggests that a few relatively stable individual characteristics (shown in gray) can influence whether employees are more likely to engage in IFBL behaviors, but characteristics such as personality and propensity are less actionable than the five readiness factors.

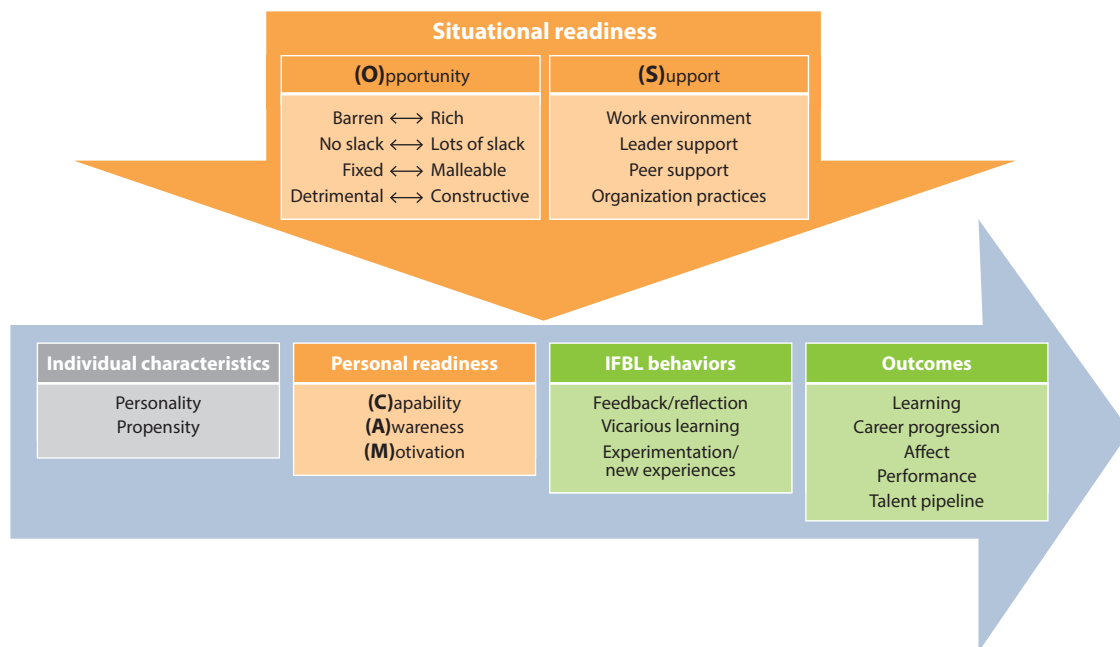


Figure 3

Organizing framework: informal field-based learning (IFBL) and Capability, Awareness, Motivation, Opportunity, and Support (CAM-OS).

The constructs described in the CAM-OS framework are not unique to IFBL. They also appear in the strategic human resource management (HRM) literature [compare with the ability, motivation, opportunity framework (Jiang et al. 2012)], the formal training literature [compare with science of training (Salas et al. 2012)], and even the education literature (compare with Ambrose et al. 2010). However, the nature and relative emphasis of the constructs differ in an IFBL context, as we highlight later.

Outcomes of Informal Field-Based Learning

Meta-analytic analyses have shown that, on average, informal learning is positively related to a wide range of desirable affective, cognitive, and performance-related outcomes (Cerasoli et al. 2018). From an affective standpoint, engaging in informal learning behaviors has been linked to increased task satisfaction (Jeon & Kim 2012), affective enjoyment and overall job satisfaction (Rowden 2002), and improved attitudes and lower job insecurity (Sanders et al. 2011). Yoon et al. (2018) found that informal learning influences organizational commitment through improved self-efficacy. Others have shown that informal learning can increase work engagement (Susomrith & Coetzer 2019, Uzunoma et al. 2021). Furthermore, informal learning can increase team members' subsequent team commitment, above that attributable to individual predispositions (Kukenberger et al. 2015).

In terms of cognitive outcomes, informal learning has been shown to increase knowledge and skill acquisition. For instance, Froehlich et al. (2014) demonstrated that engaging in informal learning can increase core skills as well as more distal perceived career development. Additionally, Matsuo & Nakahara (2013) found that informal learning is associated with increased knowledge

acquisition and knowledge sharing. From a performance standpoint, informal learning and IFBL have generally been positively linked to performance improvements (e.g., Wolfson et al. 2018, 2019). Strategically, if IFBL boosts attitudes, knowledge, and performance, it should promote the establishment of a healthy talent pipeline and a pool of ready successors (Brymer et al. 2019, Ployhart et al. 2011).

IFBL outcomes conclusion. Engaging in IFBL typically results in positive outcomes, but not always, as doing so may prohibit an employee from completing their assigned work on time, create safety risks, be perceived as overreach, or increase stress. It is important to differentiate IFBL opportunities that are constructive from those that are detrimental.

Individual Characteristics

Cerasoli et al.'s (2018) meta-analysis examined a variety of individual characteristics that could influence informal learning. They and others have found few consistent effects for employee demographics. For instance, Berg & Chyung (2008) showed a positive effect for age on informal learning, whereas Livingstone & Stowe (2007) found a negative one. Bednall et al. (2014) and Berg & Chyung (2008) found that females more readily engaged in informal learning, but Wolfson et al. (2018) found the opposite or insignificant effects. Thus far, it does not appear that surface demographics play a consistent, meaningful role in IFBL.

There are, however, at least two relatively stable personal predispositions that appear to be related to IFBL: learning goal orientation (LGO) and promotion focus. Employees with a learning goal orientation are more interested in improving than in appearing competent and are more likely to engage in IFBL behaviors (Choi & Jacobs 2011). LGO appears to affect IFBL in several ways. Matzler & Mueller (2011) found that a learning orientation positively influenced knowledge sharing, whereas a performance orientation negatively influenced it. Wong et al. (2012) suggested that individuals with a high LGO should be more likely to reflect on experiences (a key activity in informal learning) in a useful manner and learn more from them.

Regulatory focus captures how individuals self-regulate in goal pursuits and is composed of prevention focus and promotion focus (Higgins 1997). Prevention focus acts as a motivator by encouraging loss prevention, whereas promotion focus activates gain-seeking. Wolfson et al. (2018, 2019) demonstrated that promotion focus was positively related to engaging in IFBL, but prevention focus was not significantly related.

Although researchers have suggested exploring the role of personality in informal learning, in particular the personality trait of openness to experience, the evidence has been thus far inconclusive (Noe et al. 2013). However, openness to experience may indirectly lead to IFBL through other more proximal predictors such as motivation to learn, training proficiency, and participation in active learning and self-development activities (Barrick & Mount 1991, Colquitt et al. 2000, Major et al. 2006, Orvis & Leffler 2011). As propensities and personality are relatively stable in nature, they are more difficult to change than the personal readiness factors we discuss below, such as motivation. Nonetheless, it is helpful to know that although any employee may engage in IFBL under the right circumstances, some individuals may gravitate toward learning activities more readily than others due to their natural propensities.

Individual characteristics conclusion. Currently there is relatively little evidence that demographics or personality traits directly impact IFBL, but propensities such as LGO may. If an organization is keenly interested in stimulating IFBL, they might consider hiring more employees who possess these propensities, but in general, propensities are less actionable than the five readiness factors we discuss next.

Personal and Situational Readiness Factors

Although additional research is needed to understand the antecedents, dynamics, and consequences of IFBL more fully, existing research reveals some useful insights with which to guide practice in organizational settings. Although informal learning is by definition less structured and organizationally driven than formal training, that does not mean an organization or leader must “let the chips fall where they may.” Armed with what is known, IFBL can and should be encouraged, guided, and facilitated such that employees are more likely to engage in IFBL behaviors constructively, successfully, and at appropriate times. Although researchers may understandably seek to differentiate formal and informal learning, in work settings there can be a blurry or semipermeable border between them, for example, by using training, coaching, or tools to prepare employees for informal learning.

As we reviewed the theories and research related to IFBL (e.g., Cerasoli et al. 2018; Noe et al. 2014; Tannenbaum et al. 2010; Wolfson et al. 2018, 2019) and considered the nature of the workplace, we identified five essential factors for optimizing and promoting IFBL. As reflected in the CAM-OS framework, three of these factors are at the individual level, or learner related (Personal Readiness: Capability, Awareness, Motivation), and two of the factors are organizational in nature (Situational Readiness: Opportunity, Support). We describe each of these below, and then use the framework to offer actionable advice for the three stakeholder groups discussed previously: senior leadership, team leaders/managers, and individual learners.

Personal Readiness

Unlike mandatory training, engaging in informal learning is a discretionary act. By definition, a learner has more control regarding IFBL than they do in formal training, so the employee plays an oversized role in whether and how IFBL happens. We suggest that three main factors combine to establish personal readiness: capability, awareness, and motivation. Are employees capable of getting the most out of potential learning opportunities, do they know where and what to learn (and what to avoid), and are they motivated to engage in IFBL?

Capability. The first personal component that can drive IFBL is capability, which refers to competencies that enable employees to gain the most from informal learning opportunities. When placed in a new job or situation, some employees seem to readily engage in constructive IFBL behaviors and rapidly get “up to speed.” Given equal opportunities and motivation, some individuals will learn more than others. General mental ability (GMA) could play a salient role, as it has been positively linked with training outcomes (Colquitt et al. 2000); however, others have found negative relationships between GMA and informal learning and have speculated that those with higher GMA may be less likely to recognize the potential value of informal learning (Noe et al. 2013). Rather than focusing on GMA specifically, as it is unlikely to be actionable, we would suggest that one should focus on developing or teaching informal learning skills.

What types of competencies should enable an employee to be a more capable field-based learner? As IFBL often involves observing and talking with others, it should be advantageous to learn to be a more skilled observer and a better question asker. As IFBL often relies on doing something new or different, and it is difficult to learn in those instances without feedback, employees could benefit from learning how to seek, accept, and even encourage feedback from others. Finally, faster learners seem to have greater self-awareness and deal with their mistakes more constructively, so meta-cognitive and self-reflection skills (Svensson et al. 2004), along with insights about learning from errors (Katz-Navon et al. 2009), should boost an employee’s capability to learn from on-the-job experiences.

Awareness. In a formal training program, learning objectives and instructional elements are typically specified in advance and trainees are guided through a series of instructional experiences (Kraiger & Ford 2021). Trainees are all aware that they are in training, and most of them know what that experience will entail. In contrast, employees who partake in IFBL do not have a specified curriculum or a common sequence of activities they experience. Employees can have very different perceptions about the availability and utility of field-based learning opportunities and possess a different understanding about how they should approach informal learning. Although some learning can occur incidentally, intentional IFBL requires awareness on the part of the learner. Employees should be more likely to engage in constructive IFBL behaviors when they are aware of potential learning opportunities—knowing what, when, and how to learn in their work environment, including the types of opportunities that may be available or creatable. Collectively this awareness can be thought of as learning attunement. Above, we noted that although IFBL usually leads to positive outcomes, there are times when engaging in IFBL can be ineffectual or even detrimental. For that reason, employees cannot only benefit from an awareness of where and how they may be able to engage in healthy IFBL, but they should also be aware of the landmines to avoid.

Motivation. Two employees who work in the same environment are likely to engage in different amounts of informal learning because of motivational differences (Colquitt et al. 2000). In the IFBL context, motivation manifests as a willingness to find and engage in learning opportunities, to exert discretionary effort to seek opportunities, experiment, and ask for feedback. Motivation to engage in IFBL can be related to the personal predispositions we mentioned above, such as LGO (Choi & Jacobs 2011). Additionally, more curious individuals more readily engage in informal learning (Lohman 2009). But motivation to engage in IFBL is not simply a function of personality or propensities. Motivation to learn can ebb and flow more readily and is more situationally influenced than personality (Klein et al. 2006). Motivation to engage in IFBL may be boosted by encouragement from peers and leaders, targeted organizational practices, perceived psychological safety, and seeing others who actively engage in IFBL be rewarded. When employees perceive strong support for learning, they are likely to exhibit greater motivation to learn (Al-Eisa et al. 2009).

Although we have described these three personal components separately, they likely work in tandem. A deficiency in any of the three could hinder IFBL, but it is also likely that possessing two of these components could increase the likelihood of developing the third. We propose the following:

- A capable and motivated learner is more likely to actively seek out and become aware of learning opportunities.
- A capable and aware learner is more likely to be motivated to engage in IFBL.
- A motivated and aware learner is more likely to engage in IFBL behaviors, providing them with opportunities for becoming a more capable learner.

Personal readiness conclusion. Learners have more control regarding IFBL than they do in formal training, so the employee plays a bigger role in whether and how IFBL occurs. Actions that boost an employee's capabilities, awareness, and motivation are likely to enhance IFBL.

Situational Readiness

The work context or situation also influences the extent to which employees will engage in IFBL and whether doing so will yield desired outcomes. We have identified two main situational readiness factors: opportunity and support.

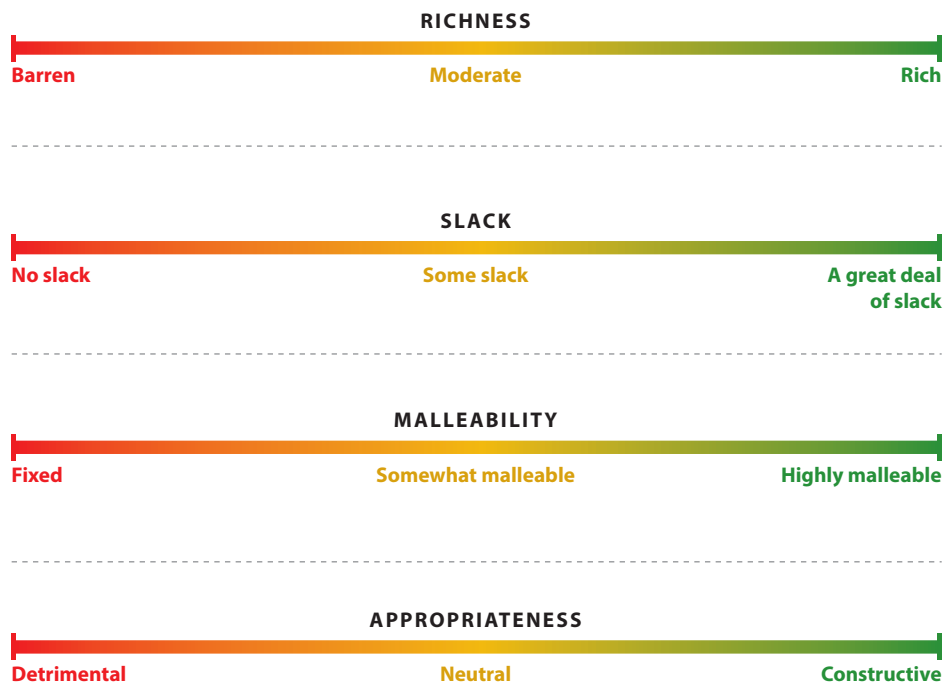


Figure 4

Four dimensions of opportunities for informal field-based learning, increasing from left to right.

Opportunity. The first organizational factor is opportunity, or the extent to which employees have the chance to engage in IFBL. Field-based learning does not happen in a vacuum. Employees engage in IFBL behaviors in a specific work environment (i.e., the field), and some settings are imbued with more natural learning opportunities than others. As **Figure 4** shows, it can be helpful to think about an employee's work environment as falling along multiple continua from learning barren to learning rich, fixed to highly malleable, no slack to a great deal of slack, and detrimental to constructive. At any point in time, a work setting can be described by where it falls along these four continua, but it is important to recognize that location on a continuum is not permanent; learning opportunities can grow, emerge, change, recede, or decline over time.

Richness. An employee may work in an environment that is rife with abundant, naturally occurring learning opportunities. For instance, some environments can offer job rotations, task variety, greater levels of interdependence among employees, and even greater levels of task complexity. These can be thought of as learning-rich environments. Other work environments contain some naturally occurring learning opportunities, but they are not as abundant, as obvious, or as continual as in a learning-rich environment. Some work environments provide little or no naturally occurring learning opportunities. All else being equal, we should expect to see higher levels of IFBL in environments that afford more learning opportunities (Cerasoli et al. 2018).

Slack. Organizational scholars have used the term slack to refer to a state in which there is more than enough resources to meet minimum needs (March & Simon 1958). Time is one such resource. Even when learning opportunities exist in theory, sometimes the work environment does not afford an employee the time to take advantage of those opportunities. For example, if an

employee's day is consistently filled with required tasks that afford little chance for learning, then that employee has little or no slack time to observe others, seek feedback, or try new tasks or approaches (Klein et al. 2006). In that instance, even if learning opportunities abound in the environment, the employee does not really have many viable opportunities for informal learning. Time availability has been shown to have a small but positive relationship with informal learning (Cerasoli et al. 2018). The effect may be small because an employee with a great deal of time might be working in a setting with limited opportunities for informal learning. In other words, a boring job with a lot of down time may have high slack but low opportunity for learning. As such, slack may have a curvilinear relationship with IFBL (Van Ruysseveldt & Van Dijke 2011).

Although some individuals may perceive a lack of time as a signal to avoid IFBL, others may view it as a stimulus to create learning opportunities and decide to work smarter rather than harder. But given equally rich learning environments, we would expect that employees with little to no discretionary time in their work schedule would be unlikely to engage in IFBL, whereas those with ample slack may take advantage of those learning opportunities, but not necessarily. Slack can also pertain to physical resources, such as equipment or supplies that may be needed to engage in IFBL. For example, if an employee needs access to a piece of equipment to engage in on-the-job experimentation, and that equipment is always being used by other employees, an opportunity for learning is lost. Insufficient access to resources can inhibit IFBL in a similar way as lack of time.

Malleability. Some work environments contain fewer naturally occurring learning opportunities than others, and some afford little slack, but that does not mean that an environment is unchangeable. For example, a manager may be able to reassign work in a way that creates a learning opportunity for an employee, or an employee might ask for time to work on a project, observe an expert in action, or try something new. Furthermore, having greater control or autonomy over the way one approaches their work can further influence engagement in informal learning (Cerasoli et al. 2018, Gijbels et al. 2012). Bruning & Campion (2018) demonstrated the value of job crafting through their role–resource approach–avoidance model of job crafting, highlighting how it can improve organizationally relevant outcomes such as improved enrichment, engagement, and decreased strain. Through varied approaches to job crafting, employees could manipulate their environment to increase learning opportunities through both their role and access to others. We can think about the extent to which modifying the work environment is feasible along a continuum as well, ranging from highly malleable (easy to create learning opportunities), to somewhat malleable, to fixed (can only rely on naturally occurring learning opportunities).

Appropriateness. As we noted above, IFBL is positively related to a range of desirable outcomes (Cerasoli et al. 2018), but we should not assume that all IFBL opportunities are created equal. Opportunities can vary in terms of their appropriateness, falling along a continuum ranging from constructive (likely to lead to positive outcomes for the learner and/or organization), to neutral (less likely to lead to positive outcomes, but not harmful), to detrimental (can lead to negative outcomes, such as career derailment, degraded customer relationships, or a safety risk).

For example, an interesting IFBL opportunity may be detrimental if taking advantage of it would distract the learner from completing other organizationally valued behaviors, such as performing one's core job responsibilities. Another example of a potentially detrimental IFBL opportunity is engaging in self-guided experimentation when the consequences of failure are significant (Katz-Navon et al. 2009). If making an error while learning could create a physical hazard or runs the risk of jeopardizing the relationship with an important customer, it is probably best not to engage in that opportunity. This does not imply that learners must avoid making errors, as mistakes

can serve as a great source of learning (Metcalf 2017). Rather, this suggests that more constructive opportunities are those in which the risks are acceptable, there is adequate support from experts, or learning can take place vicariously (e.g., by watching an expert handle a risky situation) rather through risky self-guided experimentation. Notably, in healthcare settings where consequences of errors can be particularly high, engaging in IFBL can be advantageous when undertaken in a unit with a nonpunitive climate (Wolfson et al. 2018), or in jobs specifically requiring integration of new knowledge (Wolfson et al. 2019).

An otherwise constructive opportunity may also be detrimental if it requires the active involvement of a recalcitrant other. For example, seeking feedback from, asking questions of, or attempting to observe someone who does not have the time or willingness to do so can evoke reactance that overshadows any potential positive outcomes. This suggests that in work environments where learning is typically not valued, some apparently constructive IFBL opportunities that require the support of others will need to be approached with care and in some instances, avoided. There are instances where an individual may be able to convert a potentially detrimental opportunity into a safer one, by employing more suitable IFBL behaviors. For example, Wolfson et al. (2019) found that feedback and self-reflection were effective in high time pressure jobs but had a negative effect in low time pressure jobs, whereas the reverse was true for vicarious learning.

Support

The second situational readiness factor is support for IFBL. Support can be symbolic or tangible and can manifest in the form of the work environment/climate, manager and peer behaviors, and organizational practices. Ideally, these means of support collectively provide signals and resources that encourage and enable employees to take advantage of or create constructive IFBL opportunities. The Cerasoli et al. (2018) meta-analysis found that personal support (e.g., from managers and peers), formal support (e.g., organizational practices), and informal means of support (e.g., climate) were each positively related to informal learning.

Work environment. We use the term work environment to encompass culture, climate, and job characteristics. A work environment in which learning and experimentation are valued is likely to send signals that it is acceptable to try new approaches or to ask an experienced coworker why they tackled a problem in a particular way. In environments where continuous learning is appreciated, peers are more likely to support and mentor their teammates (Eddy et al. 2005). Similarly, Noe et al. (2017) found higher levels of informal learning when employees perceive support for learning from management, their work, and the organization, what they referred to as a favorable learning climate. In contrast, if the emphasis is continually on error-free, flawless execution, that could signal that the work environment is not a place where employees should be learning. A punitive, no-error climate is likely to force people to hide their engagement in IFBL or suppress it altogether (Wolfson et al. 2018).

Psychological safety is an aspect of work climate that is highly relevant for informal learning. Psychological safety refers to a shared belief that people can be genuine and feel that it is safe to take interpersonal risks. It reflects a “sense of confidence that the team will not embarrass, reject, or punish someone for speaking up” (Edmondson 1999, p. 354). When psychological safety is strong, employees should find it easier to demonstrate informal learning behaviors, for instance, to seek feedback, ask to observe others, pose questions, voice alternative perspectives, and try out new ideas. When psychological safety is low, it inhibits learning behaviors and information sharing (Frazier et al. 2017) so it can discourage employees from admitting a need to learn, seeking input, or trying something new unless they are sure they can perform it perfectly.

Two other aspects of the work environment that can influence IFBL are role clarity and job autonomy. Interestingly, a lack of role clarity can degrade psychological safety (Frazier et al. 2017). This has implications for informal learning, as ambiguous responsibilities can lead to a team member trying something that falls outside their role. Consider what can happen when an employee tries to perform a task that is “owned” by another employee. This can evoke a negative or defensive reaction from the task owner, which can subsequently inhibit the learner from trying something new again in the future. Role clarity does not mean role rigidity. When the work environment offers employees greater autonomy (e.g., entrusted with challenging tasks, able to decide how to complete tasks), employees are more likely to feel empowered and to demonstrate greater workplace learning (Matsuo & Nakahara 2013). Greater autonomy may also heighten situational malleability and positively influence motivation to learn (Liu & Fu 2011). Furthermore, autonomy may also reflect the freedom to expand one’s role (Morgeson et al. 2005) and subsequently could facilitate or even necessitate IFBL.

Manager/leader support. Managers can greatly influence whether employees are motivated to engage in IFBL and whether demonstrating IFBL behaviors leads to positive outcomes. Managers can enhance psychological safety for informal learning by vocalizing their own learning needs, asking questions when they do not understand something, seeking input from team members from whom they can learn, and responding constructively when a team member makes a mistake. They can model that it is safe to admit one’s own learning needs and demonstrate how to help others feel comfortable doing so as well. Leaders also set a tone by establishing team norms and influencing the work climate. For instance, when teams possess a sense of empowerment and work collaboratively, they demonstrate greater informal learning (Kukenberger et al. 2015). Furthermore, Myers (2021) found that having team norms that promote learning can increase individuals’ engagement in coactive vicarious learning.

In addition to fostering psychological safety and helping establish constructive team norms, supervisors can directly stimulate informal learning by creating or highlighting IFBL opportunities. They could, for example, assign an employee to a more learning-rich environment and ensure the employee has ample free time and autonomy to take advantage of it. They can also point out the existence of potential informal opportunities, encourage an employee to reflect on a recent work experience, create chances for team members to compare notes about their experiences, and reinforce the employee’s attempts at IFBL—including attempts that did not work out as planned. They can also help an employee avert detrimental IFBL opportunities, by highlighting what is off-limits and identifying situations where an employee should avoid asking questions or experimenting on their own. If a manager consistently assigns desirable tasks to, recognizes, and promotes individuals who actively engage in IFBL, it may send a signal to all team members about the instrumentality of engaging in IFBL. It also directly creates a connection between IFBL behaviors and personally desirable outcomes.

Peer support. Peers can also influence informal learning both indirectly and directly. For example, by mocking a teammate who makes a mistake or asks a question that they deem silly, peers can adversely affect psychological safety (Frazier et al. 2017). Peers can make it very difficult for a coworker to speak up or admit a lack of knowledge, greatly inhibiting IFBL. More directly, a peer can offer to cover for a coworker, allowing their colleague time (slack) to step away from their primary work responsibilities and go observe or talk with an expert. Similarly, a peer can serve as a safety net when a coworker wants to try a new task that may be a stretch for them, converting a potentially detrimental learning opportunity into a constructive one. An experienced peer can also offer informal advice and feedback, allowing the learner to engage more safely when a learning

opportunity emerges. Many field-based learning opportunities involve coworkers as a source of learning; for example, an experienced employee might invite a less experienced colleague to watch her and ask questions.

Organization practices. Formal organizational practices related to staffing, recognition, and even formal learning can influence whether employees engage in informal learning behaviors (Cerasoli et al. 2018). Recognition has been shown to be related to informal learning (Rowden 2002). From an organizational standpoint, providing employees with greater exposure to formal learning opportunities appears to have a positive effect on informal learning (Choi & Jacobs 2011). Making ample formal training available may indicate that the organization values employee development, particularly in stronger HRM systems (Bednall & Sanders 2017). Formal training can also be targeted to boost both employee and manager readiness for IFBL. Managers could be trained to encourage and support IFBL. Similarly, training and tools can be provided to employees to equip and prepare them to be better on-the-job learners (e.g., to observe, ask questions, and reflect on their experiences more effectively).

As noted above, a manager's job promotion decisions and recognition practices can influence their team's work climate. On an organizational level, the pattern of those decisions and practices across managers also offers employees clues about whether informal learning is valued. Do active learners get recognized and rewarded, or are employees who work mistake-free the ones who typically receive promotions? Work scheduling and staffing practices can also influence IFBL. A work scheduling system that allows no time for employees to do anything but complete their required job responsibilities as assigned sends a signal that informal learning is not expected and directly restricts IFBL by eliminating slack. Similarly, staffing practices can determine the amount of time employees have to engage in IFBL. Understaffing reduces the time employees have to engage in discretionary tasks such as learning and can render experts inaccessible or unapproachable.

Informal field-based learning situational readiness conclusion. IFBL does not occur in a vacuum. The context in which employees work may be more or less conducive to IFBL. When an employee has or can create ample IFBL opportunities; has at least some time to pursue those opportunities; is aware of where and how to engage in constructive IFBL (and avoid determinantal IFBL); and is supported by their peers, their manager, and/or organizational practices, we can expect more positive IFBL outcomes.

Applying the CAM-OS Framework: Tips and Advice for Stakeholders

In **Table 1**, we describe three stakeholder perspectives, highlighting each group's unique interest in IFBL and suggesting the types of practical questions that logically emerge from their perspective. For example, from an organizational/strategic perspective, viewed through the lens of senior leadership, IFBL is relevant because it can accelerate the development of human capital resources, providing a sustainable competitive advantage. IFBL can also be appealing because it promises a more cost-effective alternative to formal training. With this in mind, senior leaders may welcome answers to questions such as where and how should we invest in IFBL, how can we balance IFBL with formal T&D, and how can we tell if we are making progress?

If we take a work group/team perspective, viewing IFBL through the lens of a manager or team leader, different interests and practical considerations emerge. Team leaders may be interested in IFBL as a way to augment formal T&D, because T&D is expensive and typically requires time away from the job. In addition, as spans of control widen, many managers have less time to personally develop individuals; they need employees to be more actively involved in their own

Table 1 Stakeholder interests and questions

Perspective	Stakeholder interest	Practical questions
Organizational/strategic: through the lens of senior leadership	To be competitive, we need depth of talent, which requires continuous learning.	Should we be investing in and promoting informal learning and if so, where and how?
	We spend a great deal on formal training.	How can we balance formal and informal learning?
	We cannot get and stay “ready” through formal training alone.	How can we assess if we are making progress?
	We need to know when and where to focus and encourage informal learning.	Should we be trying to hire people who are motivated learners?
Work groups/teams: through the lens of a team leader/manager	My team needs to continually adapt and learn for us to succeed.	How can I encourage and facilitate informal field-based learning (IFBL) within my team?
	Formal training takes time and money; I cannot rely on that alone.	How can I help my team members be better and/or faster informal learners?
	I need my team to take some responsibility for their learning.	How can I give enough autonomy but avoid detrimental informal learning?
	I need to ensure work gets done while still developing my team.	How can I assign work to promote learning while balancing other needs?
Personal/career: through the lens of an employee	I want to advance and be successful in my job and career.	How does engaging in IFBL influence how I am perceived?
	I cannot rely on formal training to address all my needs.	How can I be a better on-the-job learner?
	I am being asked to take some ownership of my development.	What are the best opportunities for me to learn?
	I am not sure what or how I am supposed to learn while on the job.	Who should I watch and talk with and how can I best learn from them?
	I do not want to get in trouble for trying things that I should not.	What should I try on my own and what should I avoid trying?

development. As such, practical considerations for managers include how best to encourage and facilitate healthy IFBL, ensure team members avoid potentially detrimental IFBL, and enable employees to become better self-guided, field-based learners.

Finally, employees have their own personal/career perspective on IFBL. Employees may be interested in IFBL as a means to advance their careers. IFBL also provides a vehicle through which employees can take ownership of their own development and focus on specific capabilities in which they are interested, and potentially uniquely craft their job to allow for personal growth. As a result, employees should want answers to the following questions: Where are the best opportunities for IFBL, who can I learn from, and how will I be viewed if I engage in IFBL?

Each stakeholder group has somewhat different needs and expectations and, as such, could apply the lessons from the research in different ways. Below are a set of practical tips for each group, based on their needs and the research literature, aligned with the CAM-OS framework (**Table 2**; also see the sidebar titled Two Techniques for Enhancing Informal Field-Based Learning).

KEY CHALLENGES AND DIRECTIONS FOR FUTURE RESEARCH

Although research has uncovered a great deal of useful information about IFBL, there is much more to be learned. Below, we highlight a few of the challenges for future IFBL research, describe several methodological needs, and offer a set of research questions for future consideration.

Table 2 Applying the CAM-OS framework: tips and advice for stakeholders

Recommendations for senior leaders	
Capability	Task your L&D function to develop training and tools that build IFBL capabilities
	Suggest that IFBL skills (e.g., what/how to observe, reflect, and ask questions) be built into existing curricula at key career points
Awareness	Highlight common field-based learning opportunities that are likely to be available at various career points
	Guide your organization's L&D function to identify common IFBL opportunities by conducting field-learning needs analyses
Motivation	Communicate about the importance of IFBL to employees
	Reinforce the message that your organization values trying, asking, and observing as ways of learning
	Provide positive feedback to managers who support IFBL
Opportunity	Ensure that the talent review process incorporates (and values) stretch assignments and other experiential learning opportunities
	Encourage managers to create more informal learning opportunities (e.g., chances to informally observe and experiment)
	Inquire whether employees have ample time to engage in IFBL
Support	Insist that your organization's practices and processes support IFBL (e.g., rewards, promotions, training, staffing)
	Allocate resources toward the development of tools and training that enable IFBL
	Take actions that promote and model psychological safety (e.g., treating mistakes as learning opportunities)
	Encourage experts to allow others to observe them, ask questions of them, and debrief with them
Recommendations for managers/team leaders	
Capability	Coach your team members to be more effective field-based learners
	Model how to reflect upon, seek feedback about, and learn from work experiences
Awareness	Boost your team's learning attunement by highlighting constructive IFBL opportunities
	Engage in one-on-one conversations with team members to surface potential IFBL opportunities for them
	Complete a red-yellow-green chart for your team
Motivation	Communicate your belief in the usefulness of IFBL
	Ask team members to share with others what they learned from an IFBL opportunity
	Treat well-intended mistakes as learning opportunities
	Ask team members what they hope to learn in the next six months
Opportunity	Assign work in a way that balances accomplishing tasks with providing learning-rich opportunities
	Connect team members with experts to stimulate future possible interactions
	Ensure team members have ample time in which to engage in IFBL
	Conduct a local field-learning needs analysis
Support	Encourage team members to support each other's IFBL efforts and acknowledge those who help others
	Boost psychological safety for learning by vocalizing your learning needs and responding constructively when others make mistakes
	Look for chances to provide team members with greater autonomy in how they do their work
Recommendations for individuals	
Capability	Build your informal learning skills and techniques
	Try to become better at asking questions, observing others, seeking feedback, reflecting on experience, and conducting debriefs
Awareness	Be alert for IFBL opportunities, including small ones that may exist in your current work assignments, as well as situations where you can create a learning opportunity
	Before taking on a potentially risky learning opportunity, ask if this is likely to be detrimental rather than constructive
Motivation	Recognize the personal and career value of engaging in IFBL
	Acknowledge that formal training is unlikely to be enough for you to become as capable and successful as you would like
	Try to be open to new experiences and willing to learn

(Continued)

Table 2 (Continued)

Recommendations for individuals	
Opportunity	Create learning opportunities by asking for stretch assignments
	Seek out experts to observe and talk with
	Consistently ask questions of experts to unpack why they do what they do
	Be careful not to take on so many learning opportunities that you are unable to complete your required work
Support	Make it easier for others to support your informal learning efforts by thanking them when they provide you with feedback, answer your questions, or let you observe them in action
	Actively solicit the support of your manager and peers for your IFBL efforts
	Offer to support your colleagues' IFBL efforts, for example, covering for them so they have time to take on an emerging learning opportunity

Abbreviations: CAM-OS, Capability, Awareness, Motivation, Opportunity, and Support; IFBL, informal field-based learning; L&D, learning and development.

Challenges and Research Implications

The fluid nature of IFBL presents several research challenges that do not typically arise when studying formal training experiences. Most training experiences have a defined beginning and end and are based on a prespecified set of desired learning objectives. IFBL does not. Similar to formal learning, IFBL researchers need to consider whether the focus of their investigation is motivation, participation, and/or outcomes. But because IFBL is often unplanned, it is difficult for a researcher to observe IFBL “in action” or to administer a test in the classic sense. These challenges have implications for future research in IFBL.

As IFBL can occur through an ongoing series of unscheduled events, event sampling methodology (Reis & Gable 2000) could be a useful research tool for understanding how IFBL unfolds over time. Similarly, future research could apply qualitative methods (e.g., diary keeping) that can yield rich descriptive information about personal and situational readiness, emergent learning opportunities and obstacles, learner behaviors, and consequences of IFBL. Because IFBL unfolds over time, with both situational and personal readiness likely to ebb and flow, future research could greatly benefit from the use of longitudinal designs (Wolfson et al. 2018, 2019). Regardless of whether the goal is to measure motivation, learning, or performance, there is the need for designs that establish baselines and capture repeated measures of KSAOs that change as a result of IFBL,

TWO TECHNIQUES FOR ENHANCING INFORMAL FIELD-BASED LEARNING

Field learning needs analysis: A field learning needs analysis (FLNA) is analogous to a training needs analysis, but rather than focusing on training gaps, it identifies on-the-job learning needs and opportunities and the ways employees can intentionally acquire competence while on the job. A FLNA typically involves interviews or focus groups with individuals familiar with the work area of interest. It explores (a) what employees could try to learn on the job; (b) the learning environment, including common learning opportunities and how best to learn from them; and (c) ways to encourage healthy IFBL practices.

Informal field-based learning red-yellow-green (RYG) chart: A red-yellow-green (RYG) chart is produced for a specific job/work environment to clarify the IFBL space for potential learners. It lists tasks a learner might try to become better at through trial and error and categorizes each as red (this is not something to learn through trial and error; full stop); yellow (only try this when accompanied by an experienced person; safety net needed); and green (safe to try; go for it). An IFBL RYG chart helps employees avoid detrimental learning opportunities and heighten their awareness of constructive opportunities.

or changes in performance that occur after controlling for prior performance and engagement in IFBL.

With the constructs in the CAM-OS model in mind, it is important to thoughtfully assess where self-report or other-report measures are preferable. For instance, the self-reflection component of IFBL and several of the personal readiness factors are almost impossible for others to observe, and thus self-report measures may be more accurate. In contrast, supervisors and peers could provide insights about their colleagues' informal feedback seeking and question asking behaviors, as well as provide perceptions about the richness and malleability of the work environment.

Future Research Directions

We have attempted to glean insights from the existing research in the context of the needs of various stakeholders. The existing research allowed us to answer several key questions, but other ones remain. In **Table 3** we offer recommendations for future research that could help enhance our understanding of IFBL and provide additional guidance to the three key stakeholder groups.

In this review we discussed a variety of antecedents, situational factors, and outcomes of IFBL, but many others may warrant consideration. For instance, Komarraju et al. (2011) explored how the Big Five personality traits are connected to learning; future research could explore the role of these traits as they pertain to IFBL. LGO has been linked to engagement in informal learning, and future research could consider the role of other forms of goal orientation such as avoid performance and prove performance goal orientation and how they independently and jointly influence IFBL. Other learner characteristics such as self-efficacy or perhaps learning attunement—a learner's readiness to learn—can influence who ultimately engages in IFBL and whether those attempts are effective.

A recurring theme here is that situational characteristics can impact who engages in and ultimately benefits from IFBL. Future researchers should consider organizational factors such as size, type, structure, and industry, as well as cross-cultural differences as potential factors facilitating or hindering IFBL. For example, is a culture with a strong history of apprenticeship programs (e.g., Germany) more amenable to IFBL? Are cultures that highly value knowledge acquisition and expertise more hospitable to employees who actively engage in informal learning behaviors? Might hierarchical cultures or ones that value deference to authority inhibit experimentation or questioning experts?

IFBL inherently involves learning from others, so the breadth or depth of coworkers' KSAOs should influence how effectively IFBL can occur. The social capital of the learner may influence whether others are willing to provide feedback or allow themselves to be observed. Furthermore, engaging in feedback seeking and vicarious learning may play a role in strengthening relationships and over time could improve social capital resources and shared experience (Wolfson & Mathieu 2021). Future research should examine how social capital enables and perhaps is developed through IFBL.

Learning often occurs in response to various events, so debriefs and after-action reviews (Lacerenza et al. 2018) can act to accelerate learning from field experiences. Future research could also consider the role of training in IFBL and whether it is possible to train individuals to be better informal learners. The use of formal training and debriefing to stimulate more effective IFBL are potential ways to bridge the semipermeable membrane between formal and informal learning.

As technology has continued to change the way people interact at work, it would be useful to study how collaboration tools and ubiquitous video conferencing influence informal learning. For example, do these technologies create new or easier opportunities for vicarious learning,

Table 3 Recommendations for future research on informal field-based learning (IFBL)

Topics	Potential research questions
Individual antecedents	When might personality be linked to constructive and/or detrimental IFBL?
	Do any other forms of goal orientation predict engagement in IFBL?
	Is learning attunement (knowing what/how to learn) related to constructive IFBL?
	What is the role of self-efficacy in motivation to engage in IFBL?
	Are the three personal readiness factors (i.e., capability, awareness, and motivation) compensatory or are all required for effective IFBL?
Organizational antecedents	Does IFBL manifest differently in organizations of different sizes, types, structures, or industries?
Unit/job characteristics	Does engaging in IFBL, particularly feedback seeking and vicarious learning, improve social capital resources and shared experience?
	How does coworker expertise influence IFBL (e.g., does breadth and depth of expertise in a unit matter)?
	How do job characteristics such as autonomy, clarity, and significance influence IFBL?
Work environment	How do IFBL opportunities ebb and flow over time? Why?
	How do the four aspects of opportunity (i.e., richness, malleability, slack, appropriateness) influence IFBL behaviors and outcomes?
Subdimensions of IFBL	When is it best to engage in different forms of IFBL behaviors (e.g., observation versus experimentation versus reflection)?
Individual outcomes	Do individuals who engage in IFBL demonstrate different performance trajectories over time?
	How do various IFBL pitfalls contribute to detrimental IFBL?
	What are the most common negative consequences of detrimental IFBL (e.g., burnout)?
	How does IFBL relate to a person's perceived career readiness and rate of advancement?
Organizational outcomes	Do organizations/units with better IFBL outperform others?
	What is the short-term and long-term relationship between IFBL and employee engagement and retention?
	Does better IFBL lead to more internal job promotions (fewer external hires)?
	Does engaging in IFBL increase breadth or depth of human capital resources?
	Does enhancing IFBL result in a stronger talent pipeline (e.g., successors)?
IFBL interventions	Can learning attunement be improved through training or coaching?
	How can we help employees avoid potentially detrimental IFBL experiences?
	How can we encourage employees to engage in constructive IFBL?
	Can supervisors be trained to support IFBL while ensuring it remains employee-driven?
	How can learners use debriefs to boost IFBL?

observation of experts, and real-time feedback? Or do they decrease physical interactions reducing the opportunity to deeply engage with, observe, and learn from experts?

As Wolfson et al. (2018, 2019) and Cerasoli et al. (2018) showed, IFBL can vary in terms of positive and negative outcomes. Part of that may be contingent on whether appropriate IFBL behaviors are evoked, and as such, future research should continue to explore the three subdimensions of IFBL. By adopting a longitudinal perspective, researchers can examine whether IFBL can have a proximal negative and distal positive outcome. Additionally, heavily engaging in IFBL may have other unintended negative consequences such as burnout, and future research may consider the dark side of IFBL and explore what causes detrimental IFBL. Some of these potential pitfalls may be associated with situational characteristics, stylistic mistakes, personal propensities, timing issues, or even an imbalance between the need to learn and to complete basic job responsibilities.

From an individual standpoint, when can IFBL enable career readiness and advancement opportunities? From a strategic human capital resource management perspective, future research

can explore whether creating enabling conditions for IFBL can improve organizational outcomes such as employee retention, internal promotion rates, and improvements in the breadth and/or depth of human capital resources.

Above, we noted that the CAM-OS framework shares elements with models of formal training. There are, however, meaningful differences in the importance of those elements and how they play out in an IFBL versus a formal training context. In formal training and education, learning objectives are specified in advance, with opportunities to learn built into the curriculum. Although on occasion a savvy corporate trainer may work with line managers to help create opportunities for trainees to subsequently use a trained skill, formal training typically ends when the “class” ends. Trainees can acquire competence in formal training even without subsequent on-the-job learning opportunities because learning experiences are embedded into the training curriculum. In contrast, IFBL relies completely on on-the-job opportunities—without those, no learning can occur. Opportunities are at the heart of IFBL, and it is often up to the learner to identify or even create them. And because the learner is responsible for identifying needs and opportunities, awareness of what and how to learn while on the job and relevant capabilities such as observation and questioning skills are far more important in IFBL than in formal learning or education.

A great deal of research has been conducted on formal training, much of it focused on how to optimize instructional design and create effective formal learning experiences. Although transfer of training is discussed in the formal training research literature, the primary focus of that research is on applying what was taught, not on fostering new, emergent learning. In organizational settings, what happens after training is typically an afterthought and is often limited to a precursory consideration of whether a specific learned skill was used on the job (i.e., proximal transfer). As a result, that literature offers few insights about how on-the-job opportunities, learner awareness and related learning capabilities, and situational support can foster new, constructive, self-directed learning. Additional research on IFBL is needed that examines the elements of the CAM-OS framework in situ, as that not only would inform how best to enhance informal learning but could also yield valuable insights to help the developers of formal training who are interested in promoting more distal transfer of trained skills.

Researchers could explore what would happen if one of the components of CAM is missing and whether they operate in a compensatory manner. They could also examine the four components of opportunity (richness, malleability, slack, and appropriateness) to better understand the prevalence and nature of informal learning opportunities and use longitudinal designs to examine how and why opportunities to learn ebb and flow. There may be an inflection point where constructive IFBL behaviors turn destructive or vice versa. Furthermore, researchers should explore how support can act as an enabling condition, whether its absence is a hindrance to IFBL, and how some learners manage to engage in IFBL despite limited support.

CONCLUSION

The majority of learning in the workplace occurs outside of classrooms and formal learning environments—it happens informally, “in the field.” Leaders, managers, and individual employees all have a vested interest in promoting healthy, constructive IFBL. When done properly, IFBL can help organizations remain competitive, work groups adapt, and employees grow and advance. To guide this, we need to understand the dynamics of IFBL, for example, how personal and situational readiness can influence whether and how IFBL unfolds. There is a growing body of research on IFBL that enabled us to describe what is known about those dynamics and to offer evidence-based tips for managers and employees. But we recognize that there is still a great deal to learn about IFBL. It is our hope that future researchers continue to explicate the IFBL space, tackling the re-

search questions we identified along with new challenges that are certain to emerge. IFBL occurs in all organizations—the role for research is to clarify how best to promote healthy, productive IFBL.

APPENDIX

Learning type	Definition
Action learning	Development to improve observable behavior in problem fields (Yorks et al. 1999)
Coactive vicarious learning	“discursive learning process where individuals (i.e., a model and learner) intentionally share and jointly process a model’s work experience(s) in interpersonal interactions to construct an emergent, situated understanding of the experience(s)” (Myers 2018, pp. 613–614)
Continuous learning	Changing behavior based on deepening and broadening of skills, knowledge, and worldviews (London & Sessa 2006)
Experiential learning	Engaging in challenging experiences, reflection, and subjective reasoning to deduce new knowledge (Kolb 1984)
Formal training and development	Programs, courses, and events developed by organizations to help employees acquire KSAOs (knowledge, skills, abilities, and other characteristics) (Noe et al. 2014)
Incidental learning	A subset of informal learning that is defined as a by-product of some other activity (Watkins & Marsick 1992)
Informal field-based learning	“engaging in intentional, self-directed behavior aimed at learning new, work oriented, and organizationally valued content outside of a formal learning program” (Wolfson et al. 2018, p. 16)
Informal learning	Intentional or incidental learning, that is not highly structured, and volitional (Marsick et al. 1999)
On-the-job training	Instruction in work setting and during work (Rothwell & Kazanas 1994)
Self-directed learning	Intentional learning that occurs formally or informally, online, or in the context of social learning (Clardy 2000)
Self-regulated learning	Affective, cognitive, and behavioral processes during learning to reach a desired goal (Sitzmann & Ely 2011)
Social learning	Occurs through observation, imitation, and reinforcement (Bandura 1962)

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